



# AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



## Test Report No. 2024 – 0478



### Agricultural Machinery – Solar-Powered Irrigation System Brgy. Guintas, Leganes, Iloilo

|                                |                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------|
| Test Report Number             | 2024 – 0478                                                                                              |
| Date of Issuance               | November 11, 2024                                                                                        |
| Valid Until                    | November 11, 2029                                                                                        |
| Agricultural Machinery         | Solar-Powered Irrigation System                                                                          |
| Type                           | Groundwater Source with Reservoir, Surface Pumpset<br>Powered by 16 Monocrystalline Photovoltaic Modules |
| Pump Brand                     | PEDROLLO                                                                                                 |
| Pump Model                     | HF 20A-N                                                                                                 |
| Photovoltaic Solar Panel Brand | AE SOLAR                                                                                                 |
| Photovoltaic Solar Panel Model | AE550MD-144                                                                                              |
| Test Requested by              | KIRSKAT VENTURE<br>Brgy. Bato Bato, Mambusao, Capiz                                                      |

### SUMMARY

| Performance Criteria                                                                       | Requirement as per<br>PNS/BAFS 324:2022 | AMTEC Test         |
|--------------------------------------------------------------------------------------------|-----------------------------------------|--------------------|
| Maximum PV Performance Ratio                                                               | 18 <sup>a</sup>                         | 19.93 <sup>b</sup> |
| The pumpset shall have a test report with pump or performance curve prior to installation. | Provided                                | Not provided       |
| Total volume discharge for whole day operation, m <sup>3</sup> , minimum                   | 48 <sup>c</sup>                         | 103.07             |

<sup>a</sup> Requirement for monocrystalline PV array.

<sup>b</sup> Data from 8:30 am to 11:30 am were not included due to shading on the panels.

<sup>c</sup> Requirement based on system design of the procuring entity.

Note: ND – No Data

## OBSERVATIONS

|                                                                                                                                                                                                                                                                                              |                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1. Operator's manual                                                                                                                                                                                                                                                                         | Provided for the controller/inverter only.                                                                                        |
| 2. Previous test reports of similar brand and model                                                                                                                                                                                                                                          | None                                                                                                                              |
| 3. PV module manufacturer                                                                                                                                                                                                                                                                    | AE Alternative Energy GmbH<br>Messerschmitring 54, 86343 Königsbrunn,<br>Germany                                                  |
| 4. Pumpset manufacturer                                                                                                                                                                                                                                                                      | PEDROLLO<br>San Bonifacio (VR), Italy                                                                                             |
| 5. Time of test                                                                                                                                                                                                                                                                              | 8:00 a.m. to 4:30 p.m.                                                                                                            |
| 6. Ease of cleaning the PV modules                                                                                                                                                                                                                                                           | The PV modules can be accessed for cleaning using a ladder or any platform.                                                       |
| 7. Ease of adjusting and repairing of parts                                                                                                                                                                                                                                                  | All parts are accessible for adjustment and repair.                                                                               |
| 8. Safety                                                                                                                                                                                                                                                                                    | Rotating parts, electrical wirings, and cables of the system are enclosed and properly sealed/coated.                             |
| 9. Failure or abnormalities that may be observed on the system during and after the operation                                                                                                                                                                                                | None                                                                                                                              |
| 10. The panels in the PV array should be installed with a rigid alloy rail made of non-corrosive material.                                                                                                                                                                                   | The PV array is installed with aluminum rails.                                                                                    |
| 11. The mounting frame structure should be made from Galvanized Iron (GI) pipes or angular bars that are either primed, hot dipped galvanized with a minimum of 5 mils, or double coated with non-corrosive paint.                                                                           | The mounting frame structure is composed of welded GI pipes, channel bars, and angular bars that were properly coated with paint. |
| 12. A uniform type and specifications of PV modules, either monocrystalline or polycrystalline, shall be used for the whole array.                                                                                                                                                           | All monocrystalline PV modules used are of uniform specifications.                                                                |
| 13. The PV modules shall be able to withstand a minimum gustiness and uplift of 180 kph. The standard degradation should be a minimum of 0.5% annually as per IEC 61215-1:2021 (Terrestrial photovoltaic (PV) modules – Design qualification and type approval – Part 1: Test requirements). | Not verified due to limited technical capabilities on-site.                                                                       |

## OBSERVATIONS (Cont'n)

|                                                                                                                                                                                                                                        |                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14. The orientation of the PV array shall be as per site assessment. It is recommended that the PV array should be facing the south based on efficiency and performance.                                                               | The PV array is facing south.                                                                                                                                       |
| 15. The angle of inclination in degrees for the PV array shall be dependent on the latitude of the location maximizing solar energy to be harvested.                                                                                   | The PV array is inclined at about 8.7°.                                                                                                                             |
| 16. Placement of the PV array shall be away from any source of shade at any time of the year. If applicable, the site of installation shall be cleared of any trees and other obstructions.                                            | A nearby tree cast shadows on the PV modules starting from 8:30 a.m. to 11:00 a.m. (Figure 5).                                                                      |
| 17. The spatial distance between each PV module should be at least 20 mm.                                                                                                                                                              | The spatial distance between each PV module is 18.95 mm.                                                                                                            |
| 18. The type, capacity of pump, suction and discharge pipe material, and other basic components shall be dependent on the design of the SPIS.                                                                                          | The pump used is a surface pumpset. GI pipe was used for the discharge and suction pipes.                                                                           |
| 19. Dry running, overheating, overloading, voltage transient, and low/high voltage input protection shall be provided for the pumpset.                                                                                                 | Low voltage input can be detected by the controller. Dry running, overheating, overloading, voltage transient, and high voltage input protection were not verified. |
| 20. A trash rack, made of non-corrosive material or painted with protective coating, shall be installed along with the pump intake. The orientation of the pump shall be dependent on the system design (e.g. submersible or surface). | Not applicable to groundwater source.                                                                                                                               |
| 21. The pumpset to be used shall be dependent on the water source and total head. Casings for groundwater sources shall be provided.                                                                                                   | The system is equipped with a surface pumpset with source from groundwater.                                                                                         |
| 22. A discharge measuring device, such as flow meter or water meter, should be included with the pumpset.                                                                                                                              | A discharge measuring device was not provided.                                                                                                                      |
| 23. The pumpset controller and the PV array controller shall be insect-proofed and weather-proofed by using double-proof box and sealants.                                                                                             | The controller/inverter is inside a steel box.                                                                                                                      |
| 24. A minimum of IP58 (splash proof) rating shall be used as per IEC 60529:2019 (Degrees of protection provided by enclosures [IP Code]).                                                                                              | Not verified due to limited technical capabilities on-site.                                                                                                         |

## OBSERVATIONS (Cont'n)

|                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25. The solar panel controller should be encased with a combiner box.                                                                                                                                                                                                                                                                                                   | The controller/inverter is installed inside a combiner box.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 26. The capacity of the inverter shall be at least 25% higher than the pumpset input power requirement.                                                                                                                                                                                                                                                                 | The maximum output power of the inverter (5.5 kW) is 37.5 % higher than the pumpset's input power requirement (4 kW).                                                                                                                                                                                                                                                                                                                                                                               |
| 27. The controller shall be fully automated based on manufacturers' specifications.                                                                                                                                                                                                                                                                                     | The controller/inverter has a digital display and is fully automated.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 28. Cables and wirings used for the SPIS shall be in accordance with the Philippine Electrical Code (PEC), using PV cables specific for PV modules. Wiring installations shall be properly protected for weather conditions and other external factors.                                                                                                                 | Electrical wirings and cables of the system are enclosed and sealed/coated. However, due to limited technical capabilities on site, it was not verified if the cables and wiring used for the SPIS are in accordance with the Philippine Electrical Code (PEC) and if the wiring installations are properly protected from weather conditions and other external factors.                                                                                                                           |
| 29. If installed, the design of the reservoir should indicate that the intake be at least equal to the outflow during operation. The reservoir shall be able to withstand the design load. The water storage should also include a ladder, drain and overflow sensor, a freeboard, and intake and outflow valves. Construction shall be in accordance with the PD 1096. | <ul style="list-style-type: none"> <li>a. A 1000-liter tank is installed in the system. The inflow valve (63.5 mm) is bigger than the outflow valve (25.4 mm).</li> <li>b. The reservoir was able to withstand the load when it was at full capacity. The tank is provided with a ladder, drain, and intake and outflow valves.</li> <li>c. It was not verified if the construction of the reservoir conformed in accordance with PD 1096 due to limited technical capabilities on site.</li> </ul> |
| 30. A Rapid Shutdown Device (RSD) shall be installed for roof-mounted panels.                                                                                                                                                                                                                                                                                           | Not applicable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 31. Lightning arrester shall be installed in accordance with Article 2.80 (Surge Arresters) of the National Electrical Code (NEC).                                                                                                                                                                                                                                      | Not provided                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## OBSERVATIONS (Cont'n)

|                                                                                      |                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 32. Warning signs shall be installed for electrical components on-site.              | Warning notices are provided on the PV modules, controller, and pumpset.                                                                                                                                                             |
| 33. Surge Protection Devices (SPD) shall be installed along with the PV module.      | A surge protection device is provided.                                                                                                                                                                                               |
| 34. Personal Protective Equipment (PPE) shall be provided for each operator on site. | The operator was not provided with personal protective equipment during the test operation.                                                                                                                                          |
| 35. Other remarks                                                                    | a. Markings and labeling are provided for the PV modules, inverter, and pumpset.<br>b. The Test applicant insisted on performing the test even with intermittent pumpset discharge due to insufficient water supply from the source. |

**PURPOSE AND SCOPE OF TEST**

|                  |                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose          | Performance Testing                                                                                                                                                            |
| Date of Test     | September 30, 2024                                                                                                                                                             |
| Location of Test | Brgy. Guintas, Leganes, Iloilo<br>10°46'23.80" N, 122°35'38.80" E                                                                                                              |
| Items Tested     | Solar Panel Power Output, Input Power to the Pumpset, Pumpset Output Power, Discharge Capacity, Total Head, PV Performance Ratio, PV Module Temperature, and System Efficiency |

**METHODS OF TEST**

The Solar-Powered Irrigation System was tested based on PNS/BAFS 325:2022 Solar-Powered Irrigation System – Methods of Test.



**DESCRIPTION OF THE SYSTEM**

Figure 1. Different components of the Solar-Powered Irrigation System installed at Brgy. Guintas, Leganes, Iloilo include the following: (a) Controller house, (b) Discharge tank, and (c) PV modules.

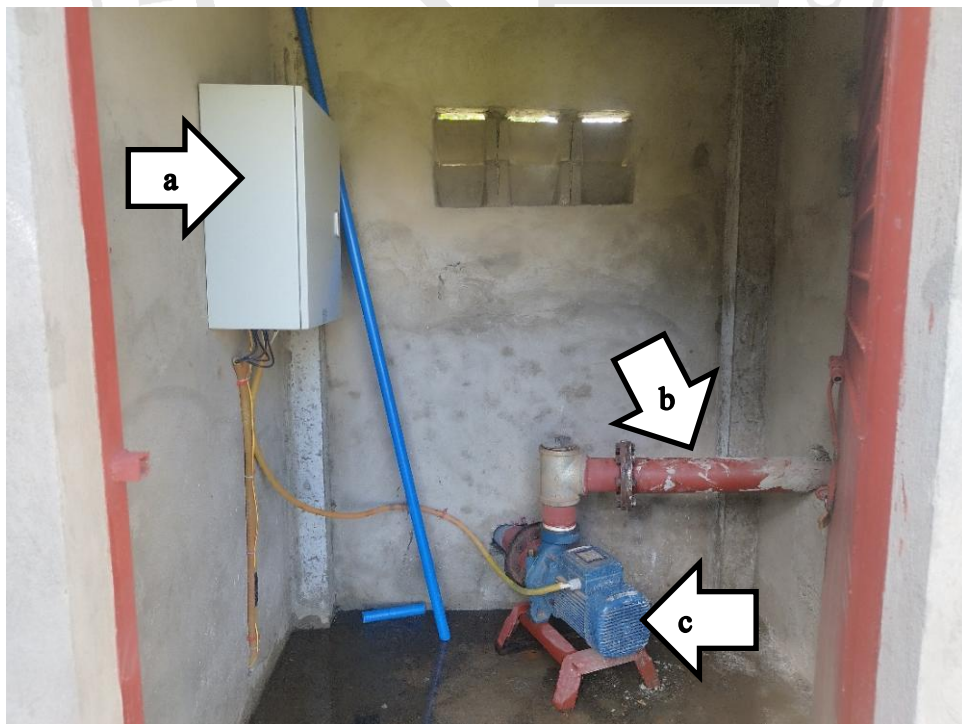


Figure 2. Different components of the Solar-Powered Irrigation System installed at Brgy. Guintas, Leganes, Iloilo include the following: (a) Controller box, (b) Discharge pipe, (c) Surface pumpset.

### DESCRIPTION OF THE SYSTEM (Cont'n)

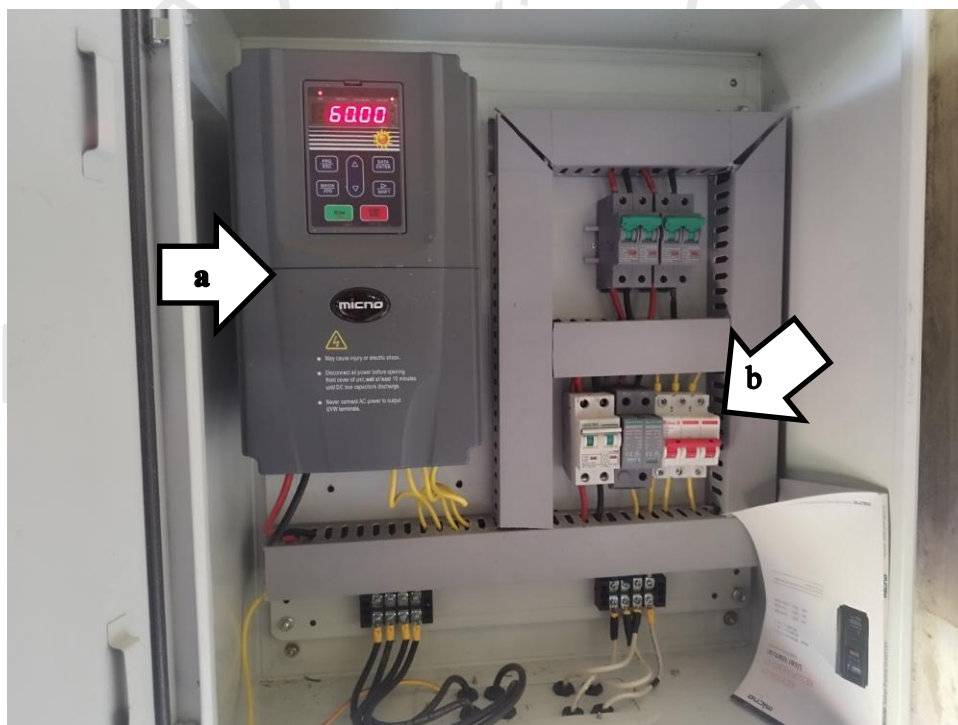


Figure 3. Different components of the Solar-Powered Irrigation System installed at Brgy. Guintas, Leganes, Iloilo include the following: (a) Controller/inverter and (b) Circuit breakers and surge protection device.



**SPECIFICATIONS**

| Item                                                                               | Manufacturer's Specification <sup>a</sup> | AMTEC Verification                     |
|------------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------|
| 1 Reservoir                                                                        |                                           |                                        |
| 1.1 Overall dimensions, mm                                                         |                                           |                                        |
| 1.1.1 Diameter                                                                     | ND                                        | 982                                    |
| 1.1.2 Height                                                                       | ND                                        | 1440                                   |
| 1.1.3 Height from the discharge outlet of the tank to the distribution channel, mm | ND                                        | NA                                     |
| 1.1.4 Thickness, mm                                                                | ND                                        | NM <sup>b</sup>                        |
| 2 Pumpset                                                                          |                                           |                                        |
| 2.1 Pump                                                                           |                                           |                                        |
| 2.1.1 Brand                                                                        | PEDROLLO                                  | PEDROLLO                               |
| 2.1.2 Model                                                                        | HF 20A-N                                  | HF 20A-N                               |
| 2.1.3 Serial number                                                                | 22A5098                                   | 22A5098                                |
| 2.1.4 Type                                                                         | ND                                        | Surface, centrifugal, Non-self-priming |
| 2.1.5 Size, mm                                                                     | ND                                        | 103 × 103                              |
| 2.1.6 Maximum discharge, m <sup>3</sup> /h                                         | 400-1700 L/min                            | 102                                    |
| 2.1.7 Maximum total head, m                                                        | 21.5-10                                   | 21.5                                   |
| 2.1.8 Manufacturer                                                                 | PEDROLLO                                  | PEDROLLO                               |
| 2.1.9 Country of manufacture                                                       | Italy                                     | Italy                                  |
| 2.2 Electric motor                                                                 |                                           |                                        |
| 2.2.1 Brand                                                                        | PEDROLLO                                  | PEDROLLO                               |
| 2.2.2 Model                                                                        | ND                                        | ND <sup>c</sup>                        |
| 2.2.3 Type                                                                         | Three-phase                               | Induction                              |
| 2.2.4 Rated power, kW                                                              | 4                                         | 4                                      |
| 2.2.5 Rated voltage, V                                                             | 220/440                                   | 220/440 <sup>a</sup>                   |
| 2.2.6 Rated current, A                                                             | 17.3/10.3                                 | 17.3/10.3 <sup>a</sup>                 |
| 2.2.7 Rated frequency, Hz                                                          | 60                                        | 60 <sup>a</sup>                        |

<sup>a</sup> The data were obtained from the component's nameplate.

<sup>b</sup> Not accessible for measurement.

<sup>c</sup> Not provided on the component nameplate.

Note: NA – Not Applicable (The distribution channel was not installed at the time of test)

ND – No Data, NM – Not Measured

**SPECIFICATIONS (Cont'n)**

| Item  |                             | Manufacturer's Specification <sup>a</sup> | AMTEC Verification         |
|-------|-----------------------------|-------------------------------------------|----------------------------|
| 3     | Controller/Inverter         |                                           |                            |
| 3.1   | Brand                       | MICNO                                     | MICNO                      |
| 3.2   | Model                       | AE300-01-5R5G-S2                          | AE300-01-5R5G-S2           |
| 3.3   | Enclosure class             | ND                                        | ND <sup>b</sup>            |
| 3.4   | Input                       |                                           |                            |
| 3.4.1 | Maximum power, kW           | ND                                        | ND <sup>b</sup>            |
| 3.4.2 | Maximum input voltage, DCV  | 440                                       | 440 <sup>a</sup>           |
| 3.4.3 | Maximum input current, DCA  | ND                                        | ND <sup>b</sup>            |
| 3.5   | Output                      |                                           |                            |
| 3.5.1 | Maximum power, kW           | 5.5                                       | 5.5 <sup>a</sup>           |
| 3.5.2 | Maximum output voltage, ACV | 25                                        | 25 <sup>a</sup>            |
| 3.5.3 | Maximum output current, A   | 220                                       | 220 <sup>a</sup>           |
| 4     | Solar module                |                                           |                            |
| 4.1   | Brand                       | AE SOLAR                                  | AE SOLAR                   |
| 4.2   | Model                       | AE550MD-144                               | AE550MD-144                |
| 4.3   | Manufacturer                | AE Alternative Energy GmbH                | AE Alternative Energy GmbH |
| 4.4   | Country of manufacture      | Germany                                   | Germany                    |
| 4.5   | Cell type                   | ND                                        | Monocrystalline            |
| 4.6   | Cell size, L × W, mm        | ND                                        | 181.23 × 90.16             |
| 4.7   | Cell configuration          | ND                                        | 24 × 6                     |

<sup>a</sup> The data were obtained from the components' nameplate.

<sup>b</sup> Not provided on the component's nameplate.

Note: ND – No Data

**SPECIFICATIONS (Cont'n)**

| Item                                                     | Manufacturer's Specification <sup>a</sup> | AMTEC Verification                          |
|----------------------------------------------------------|-------------------------------------------|---------------------------------------------|
| 4.8 PV module dimension, mm                              |                                           |                                             |
| 4.8.1 Length                                             | ND                                        | 2272                                        |
| 4.8.2 Width                                              | ND                                        | 1134                                        |
| 4.9 Total number of solar modules                        | ND                                        | 16                                          |
| 4.9.1 Number of solar modules per string                 | ND                                        | 8                                           |
| 4.9.2 Number of strings                                  | ND                                        | 2                                           |
| 4.10 Nominal power, $W_p$ , W                            | 550                                       | 550 <sup>a</sup>                            |
| 4.11 Short circuit current, $I_{sc}$ , A                 | 13.67                                     | 13.67 <sup>a</sup>                          |
| 4.12 Open circuit voltage, $V_{oc}$ , V                  | 51.44                                     | 51.44 <sup>a</sup>                          |
| 4.13 Current at maximum power, $I_{mpp}$ , A             | 12.92                                     | 12.92 <sup>a</sup>                          |
| 4.14 Voltage at maximum power, $V_{mpp}$ , V             | 42.57                                     | 42.57 <sup>a</sup>                          |
| 4.15 Reference standard temperature condition, $T_{STC}$ | 45±2                                      | 45±2 <sup>a</sup>                           |
| 4.16 Orientation                                         | ND                                        | South                                       |
| 4.17 Angle of inclination, °                             | ND                                        | 8.7                                         |
| 5 Safety devices                                         | ND                                        | Surge protection device and circuit breaker |
| 6 Special features                                       | ND                                        | None                                        |

<sup>a</sup> The data were obtained from the component's nameplate.

Note: ND – No Data

## RESULTS

Table 1. Performance of the Solar-Powered Irrigation System installed in Brgy. Guintas, Leganes, Iloilo.

| Time of the Day | Solar Irradiance, W/m <sup>2</sup> | Solar Power, W | Module Temperature, °C | Ambient Conditions |                      | Solar Panel Output (DC) |      |          | Input to the Pumpset (AC) |      |          | Pumpset Output |                                       |                 | PV Performance Ratio | System Efficiency, % |
|-----------------|------------------------------------|----------------|------------------------|--------------------|----------------------|-------------------------|------|----------|---------------------------|------|----------|----------------|---------------------------------------|-----------------|----------------------|----------------------|
|                 |                                    |                |                        | Temperature, °C    | Relative Humidity, % | V                       | A    | Power, W | V                         | A    | Power, W | Total Head, m  | Discharge Capacity, m <sup>3</sup> /h | Output Power, W |                      |                      |
| 8:00 a.m.       | 610.9                              | 25183.23       | 37.3                   | 29.3               | 89.4                 | 329.0                   | 10.9 | 3585.56  | 209.9                     | 11.3 | 2371.87  | 6.98           | 12.23                                 | 232.50          | 10.62                | 0.923                |
| 8:30 a.m.       | 671.7                              | 27689.60       | 43.5                   | 29.5               | 86.0                 | 335.3                   | 10.5 | 3520.65  | 210.6                     | 11.2 | 2358.72  | 7.00           | 12.34                                 | 235.32          | 11.74                | 0.850                |
| 9:00 a.m.       | 726.9                              | 29963.75       | 42.5                   | 30.5               | 83.7                 | 359.6                   | 9.8  | 3523.59  | 211.4                     | 10.5 | 2219.70  | 6.84           | 11.58                                 | 215.78          | 13.50                | 0.720                |
| 9:30 a.m.       | 929.6                              | 38319.68       | 42.5                   | 31.4               | 76.5                 | 354.3                   | 10.1 | 3578.09  | 210.7                     | 11.3 | 2380.91  | 7.00           | 12.33                                 | 234.95          | 16.09                | 0.613                |
| 10:00 a.m.      | 1105.8                             | 45583.21       | 46.1                   | 31.2               | 77.4                 | 323.5                   | 10.6 | 3429.10  | 210.6                     | 10.3 | 2169.18  | 7.34           | 13.82                                 | 276.18          | 21.01                | 0.606                |
| 10:30 a.m.      | 965.0                              | 39778.98       | 50.6                   | 31.0               | 79.1                 | 353.7                   | 9.8  | 3466.59  | 210.8                     | 11.1 | 2339.88  | 7.54           | 14.62                                 | 300.05          | 17.00                | 0.754                |
| 11:00 a.m.      | 1092.1                             | 45021.20       | 49.8                   | 31.5               | 72.7                 | 343.3                   | 10.7 | 3673.67  | 210.8                     | 11.1 | 2339.88  | 7.45           | 14.30                                 | 290.28          | 19.24                | 0.645                |
| 11:30 a.m.      | 1012.5                             | 41737.08       | 54.1                   | 31.8               | 69.6                 | 341.9                   | 10.5 | 3589.43  | 210.8                     | 10.6 | 2234.48  | 6.70           | 10.84                                 | 197.73          | 18.68                | 0.474                |
| 12:00 m.        | 1040.5                             | 42894.08       | 53.4                   | 32.2               | 71.2                 | 345.7                   | 10.7 | 3698.46  | 211.1                     | 11.1 | 2343.21  | 6.86           | 11.68                                 | 218.23          | 18.31                | 0.509                |
| 12:30 p.m.      | 1111.0                             | 45800.31       | 52.9                   | 32.5               | 71.3                 | 351.0                   | 10.6 | 3720.25  | 210.8                     | 10.9 | 2297.72  | 6.78           | 11.25                                 | 207.74          | 19.93                | 0.454                |
| 1:00 p.m.       | 1064.2                             | 43871.07       | 51.3                   | 33.2               | 72.6                 | 353.3                   | 10.6 | 3745.33  | 211.1                     | 11.3 | 2385.43  | 6.70           | 10.84                                 | 197.63          | 18.39                | 0.450                |
| 1:30 p.m.       | 935.7                              | 38572.52       | 52.0                   | 32.0               | 69.4                 | 343.3                   | 10.0 | 3433.33  | 211.0                     | 10.3 | 2173.30  | 6.73           | 11.03                                 | 202.25          | 17.75                | 0.524                |
| 2:00 p.m.       | 843.9                              | 34788.23       | 48.8                   | 32.0               | 69.2                 | 350.3                   | 10.5 | 3677.80  | 211.1                     | 11.3 | 2385.43  | 6.98           | 12.22                                 | 232.23          | 14.58                | 0.668                |
| 2:30 p.m.       | 749.7                              | 30906.38       | 47.7                   | 34.1               | 67.1                 | 346.2                   | 10.9 | 3773.22  | 211.1                     | 10.6 | 2237.66  | 6.87           | 11.72                                 | 219.23          | 13.81                | 0.709                |
| 3:00 p.m.       | 646.4                              | 26646.66       | 43.9                   | 32.6               | 68.1                 | 351.4                   | 10.8 | 3795.48  | 210.1                     | 10.9 | 2290.09  | 7.23           | 13.36                                 | 262.91          | 11.64                | 0.987                |
| 3:30 p.m.       | 529.8                              | 21840.03       | 40.0                   | 31.6               | 69.2                 | 344.8                   | 9.8  | 3379.37  | 211.6                     | 10.4 | 2200.64  | 6.72           | 10.93                                 | 199.93          | 9.92                 | 0.915                |
| 4:00 p.m.       | 409.7                              | 16889.13       | 37.6                   | 32.0               | 70.0                 | 322.8                   | 10.2 | 3292.22  | 210.7                     | 9.4  | 1980.58  | 6.92           | 11.96                                 | 225.52          | 8.53                 | 1.34                 |
| 4:30 p.m.       | 287.4                              | 11847.54       | 35.2                   | 32.1               | 72.0                 | 310.6                   | 7.2  | 2236.56  | 148.1                     | 8.5  | 1258.85  | 6.61           | 10.39                                 | 187.09          | 9.41                 | 1.58                 |



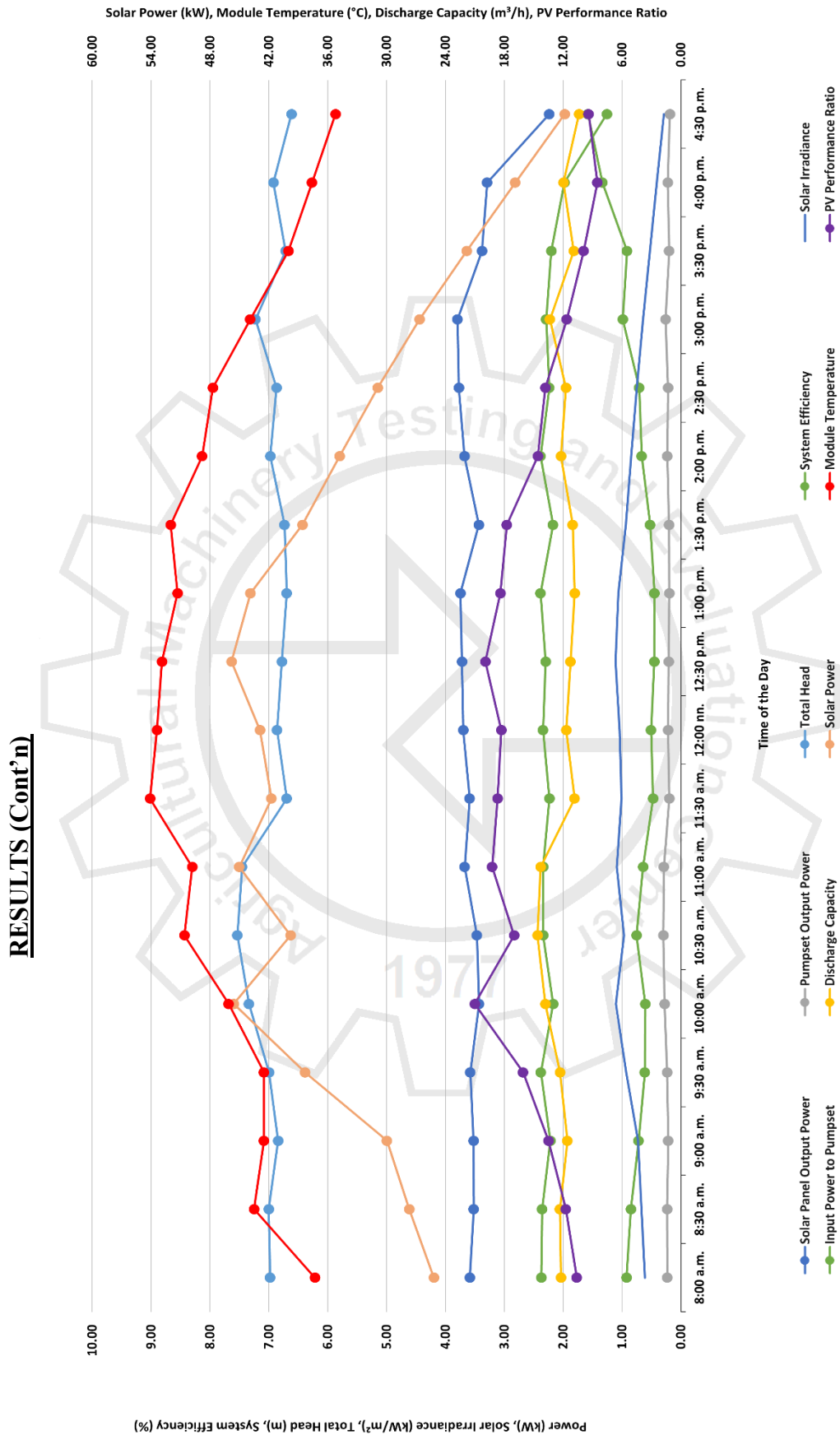


Figure 4. Performance curves of the Solar-Powered Irrigation System installed in Brgy. Guintas, Leganes, Iloilo.

### OBSERVATIONS



Figure 5. PV module shading due to nearby trees at 8:30 a.m.



Figure 6. Nameplate of the surface pumpset of the SPIS installed at Brgy. Guintas, Leganes, Iloilo.

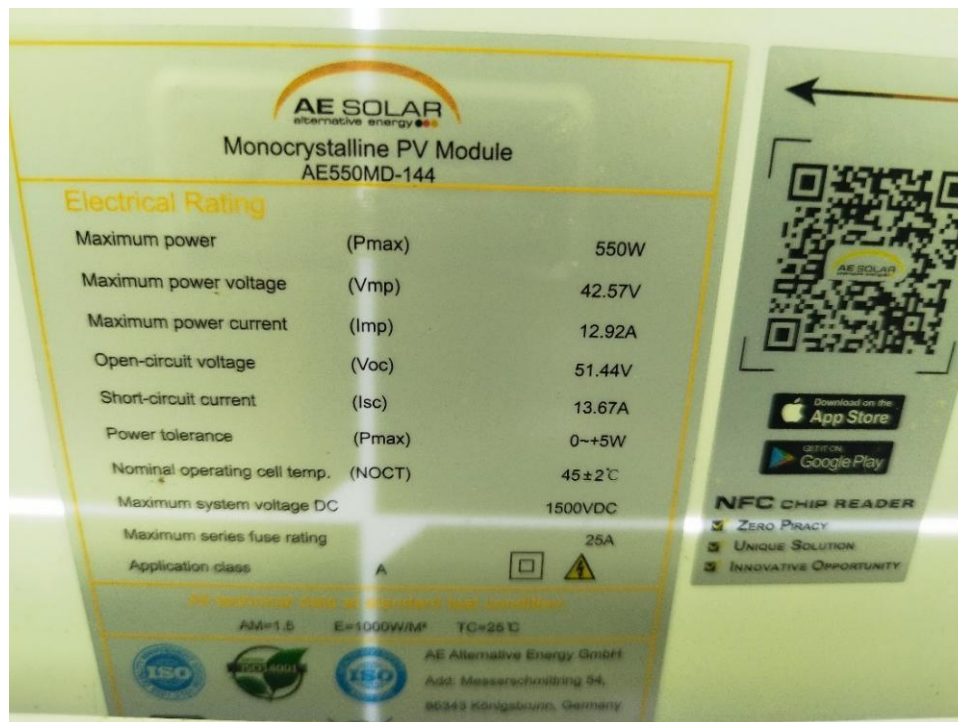
**OBSERVATIONS (Cont'n)**

Figure 7. Nameplate of the PV module of the SPIS installed at Brgy. Guintas, Leganes, Iloilo.



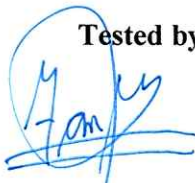
Figure 8. Nameplate of the Controller/inverter of the SPIS installed at Brgy. Guintas, Leganes, Iloilo.



**OBSERVATIONS (Cont'n)**

Figure 9. Marking and safety warnings on the components of the SPIS installed at Brgy. Guintas, Leganes, Iloilo.

Tested by:

**FRANKLIN C. PAJARON**

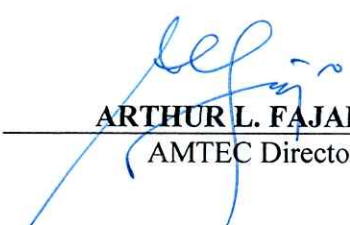
Test Engineer

Verified by:

**RHEY MARC C. ALBALOS**

Test Engineer

Approved for Release:

  
**ARTHUR L. FAJARDO**

AMTEC Director

11 NOV 2024

Date

## REMINDER

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